

Scale SEAL Leaderboards

Frontier Model Performance Across 20+ Benchmarks

Comparative analysis of 100+ models from OpenAI, Anthropic, Google, Meta, and emerging entrants — evaluated on Scale Labs' private + open-source benchmark suite spanning Agentic, Frontier, Safety & Multimodal.

20+

Benchmarks

100+

Models Evaluated

4

Capability Domains

20+ Benchmarks Spanning Four Capability Domains Evaluate 100+ Models

Scale Labs combines private datasets (anti-overfit) with open-source datasets for comparability

AGENTIC	FRONTIER	SAFETY	MULTIMODAL
Coding & Tool Use	Reasoning & Knowledge	Alignment & Honesty	Audio & Vision-Language
SWE Atlas – Test Writing	Humanity's Last Exam	MASK (Honesty)	AudioMultiChallenge (Text)
SWE Atlas – Codebase QnA	HLE (Text Only)	PropensityBench	AudioMultiChallenge (Audio)
MCP Atlas (Tool Use)	EnigmaEval	Fortress (Nat'l Security)	VISTA (Vision-Language)
SWE-Bench Pro (Public)	MultiChallenge	Prof. Reasoning – Legal	VisualToolBench
SWE-Bench Pro (Private)	MultiNRC (Multilingual)	Prof. Reasoning – Finance	
	SciPredict	Remote Labor Index (RLI)	

GPT-5.4 Dominates Agentic Coding: Top Scores on SWE Atlas & SWE-Bench Pro

Top-3 models per benchmark | scores with $\pm 95\%$ CI | private repos test generalization beyond public leaderboards

Benchmark	#1 Model (Score)	#2 Model (Score)	#3 Model (Score)
SWE Atlas Test Writing	OpenAI GPT-5.4-xHigh (Codex CLI) 44.4	OpenAI GPT-5.4-xHigh (Mini-SWE) 40.0	OpenAI GPT-5.3-Codex-Xhigh 39.0
SWE Atlas Codabase QnA	OpenAI GPT 5.4 xHigh (Codex) 40.0	OpenAI GPT 5.4 xHigh (Mini-SWE) 36.2	Anthropic Claude Opus 4.6 (Claude Code) 33.2
MCP Atlas Tool Use	Muse Muse Spark 78.2	Anthropic Claude Opus 4.6 75.0	Google Gemini 3.1 Pro Preview 73.0
SWE-Bench Pro (Public)	OpenAI GPT-5.4-pro (xHigh) 59.1	Muse Muse Spark 55.0	Anthropic Claude Opus 4.6 51.0
SWE-Bench Pro (Private)	Anthropic Claude Opus 4.6 47.1	Muse Muse Spark 44.7	OpenAI GPT-5.4-pro (xHigh) 43.4

Lab:

OpenAI

Google

Anthropic

Muse Spark

GPT-5.4-Pro Leads Humanity's Last Exam at 44.3% — Scores Remain Below 50%

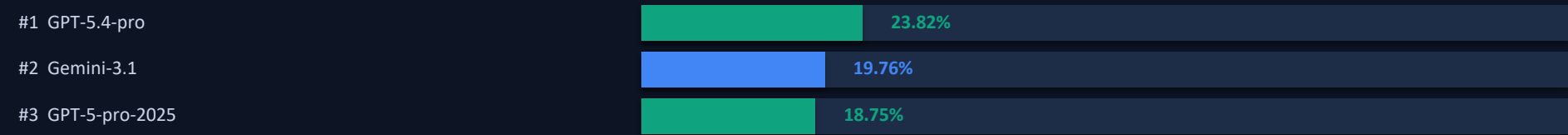
Frontier reasoning benchmarks remain highly unsaturated: even top models fall well below ceiling

Humanity's Last Exam (HLE)

← Far from 100% ceiling



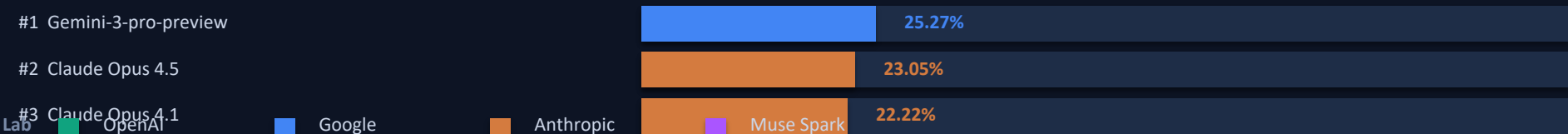
EnigmaEval



MultiChallenge



SciPredict



Lab Google

Anthropic

Muse Spark

Key insight: HLE top score = 44.3% | EnigmaEval top = 23.8% | SciPredict top = 25.3% — these benchmarks show substantial room for improvement

Gemini Models Sweep Audio and Vision-Language Benchmarks

Google's Gemini variants hold top positions across every audio and vision-language leaderboard

AudioMultiChallenge (Text Output)

All top-3: Gemini

#1 Gemini-3-pro-preview



#2 Gemini-2.5-pro



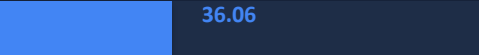
#3 Gemini-2.5-flash



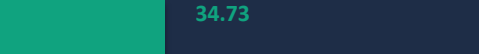
AudioMultiChallenge (Audio Output)

Gemini leads audio generation

#1 Gemini-3.1-flash-live-preview



#2 GPT-realtime-1.5



VISTA (Vision-Language)

Gemini & GPT-5.4 neck-and-neck

#1 Gemini 2.5 Pro Exp.



#2 GPT-5.4-pro



VisualToolBench

GPT-5.4 edges Gemini-3.1

#1 GPT-5.4



#2 Gemini-3.1-pro



Google Gemini holds all top-3 positions on AudioMultiChallenge (Text Output) and leads VISTA vision-language with Gemini 2.5 Pro Exp. (54.65). GPT-5.4 edges Gemini only on VisualToolBench (29.17 vs 28.97).

Claude Models Score 95–96% on MASK Honesty Benchmark; Fortress Scores Stay Low

Safety alignment varies dramatically by task type — near-ceiling honesty vs. low national-security refusal scores

STRONG ALIGNMENT PERFORMANCE

Claude leads MASK honesty & professional reasoning



CHALLENGING SAFETY BENCHMARKS

Fortress national-security scores: 8–13 across all models



Alignment performance spans 8 to 96 — no model is uniformly safe across all task types

OpenAI, Google, and Anthropic Each Claim #1 Across Different Domains

Heatmap: which lab's model ranks #1 on each benchmark — no single lab dominates everywhere

Benchmark	Domain	#1 Lab	#1 Model	Score
SWE Atlas – Test Writing	Agentic	OpenAI	GPT-5.4-xHigh (Codex CLI)	44.4
SWE Atlas – Codebase QnA	Agentic	OpenAI	GPT 5.4 xHigh (Codex)	40.8
MCP Atlas (Tool Use)	Agentic	Muse	Muse Spark	78.3
SWE-Bench Pro (Public)	Agentic	OpenAI	GPT-5.4-pro (xHigh)	59.1
SWE-Bench Pro (Private)	Agentic	Anthropic	Claude Opus 4.6	47.1
Humanity's Last Exam	Frontier	OpenAI	GPT-5.4-pro	44.3
EnigmaEval	Frontier	OpenAI	GPT-5.4-pro	23.8
MultiChallenge	Frontier	Muse	Muse Spark	75.5
MultiNRC (Multilingual)	Frontier	OpenAI	GPT-5-pro	65.2
SciPredict	Frontier	Google	Gemini-3-pro-preview	25.3
MASK (Honesty)	Safety	Anthropic	Claude Opus 4.6	96.3
Prof. Reasoning – Legal	Safety	Anthropic	Claude Opus 4.6	52.3
Prof. Reasoning – Finance	Safety	Anthropic	Claude Opus 4.6	53.3
AudioMultiChallenge (Text Out)	Multimodal	Google	Gemini-3-pro-preview	54.6
AudioMultiChallenge (Audio Out)	Multimodal	Google	Gemini-3.1-flash-live	36.1
VISTA (Vision-Language)	Multimodal	Google	Gemini 2.5 Pro Exp.	54.6
VisualToolBench	Multimodal	OpenAI	GPT-5.4	29.2

Domain wins:

OpenAI leads: 6

Anthropic leads: 4

Google leads: 4

Muse leads: 2

Key Takeaways: No Single Model Wins Everything — Specialization Is the New Frontier

01 **OpenAI** **GPT-5.4 leads agentic coding & frontier reasoning — but not multimodal**

Top on SWE Atlas, SWE-Bench Pro (Public), Humanity's Last Exam (44.32%), and EnigmaEval (23.82%). Loses to Gemini on audio/vision and to Claude on safety.

02 **Google** **Gemini dominates audio and vision-language**

Holds all top-3 positions on AudioMultiChallenge (Text Output): 54.65 / 46.90 / 40.04. Leads VISTA at 54.65. Trails on agentic coding and safety benchmarks.

03 **Anthropic** **Claude leads safety, honesty & professional reasoning**

MASK honesty: 96.28% (Opus 4.6) and 96.13% (Sonnet 4) — near-ceiling. #1 on Professional Reasoning for both Legal (52.27%) and Finance (53.28%).

04 **Insight** **Key benchmarks remain far below saturation**

HLE top = 44.3% | EnigmaEval top = 23.8% | SciPredict top = 25.3% — frontier models still have substantial room for improvement on multi-step reasoning and scientific forecasting.

05 **Muse Spark** **New entrant Muse Spark is a strong competitor across the board**

Top-3 on 6+ benchmarks: #1 MultiChallenge (75.52%), #1 MCP Atlas (78.30%), #2 HLE, #2 SWE-Bench Pro (Public & Private). A notable challenger to established labs.